

❖ Problem Statement:

“Finding the Right BIS Standard is Difficult”

- BIS standards are stored in **large, complex technical documents**
- Manual search is **time-consuming and error-prone**
- Requires **domain expertise to identify the correct standard**
- No system exists for **intelligent recommendation**

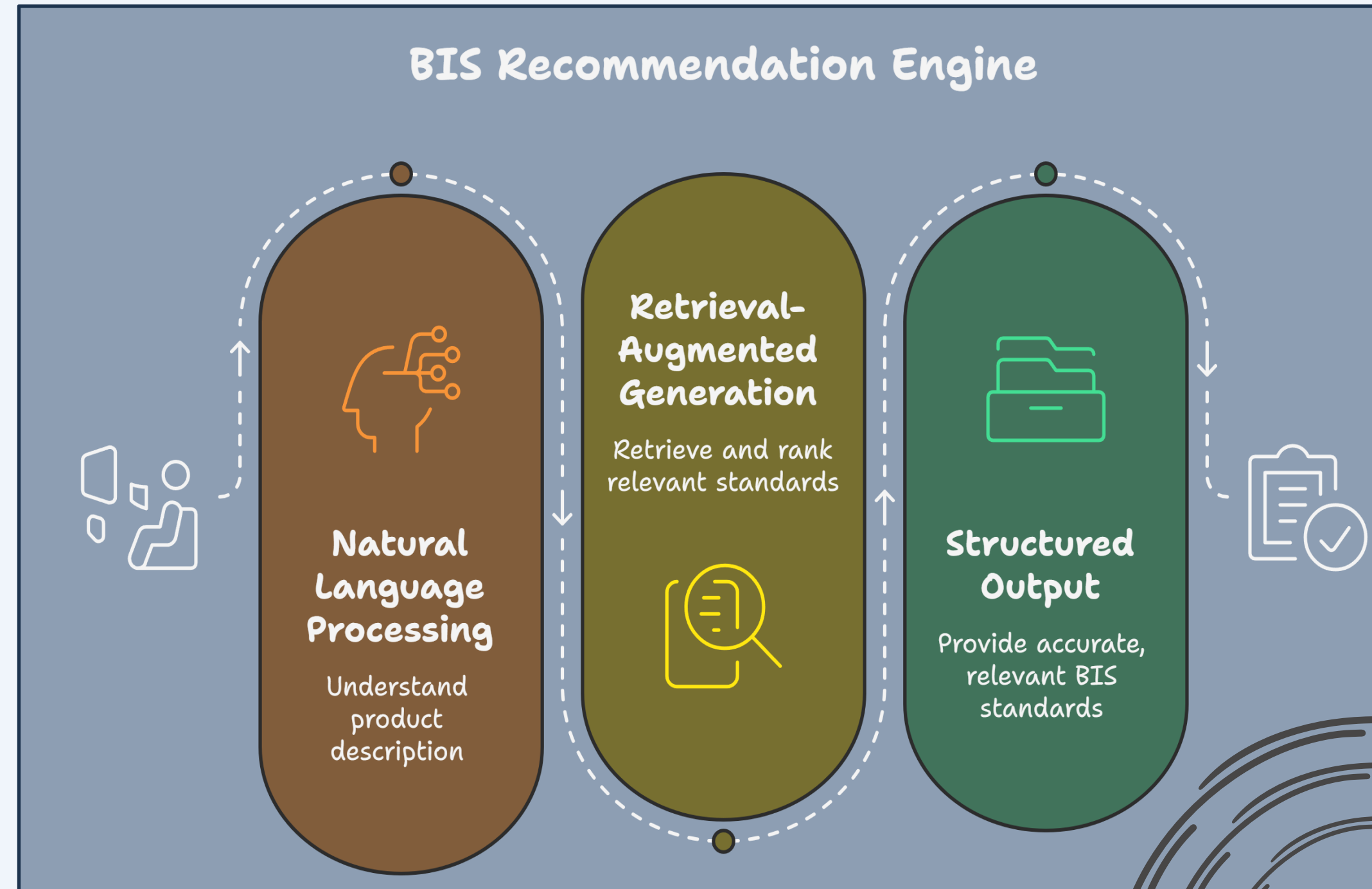
❖ Solution Overview :

❖ Key Features :

- Hybrid retrieval (semantic + keyword search)
- Intelligent reranking for precision
- Fast and scalable system

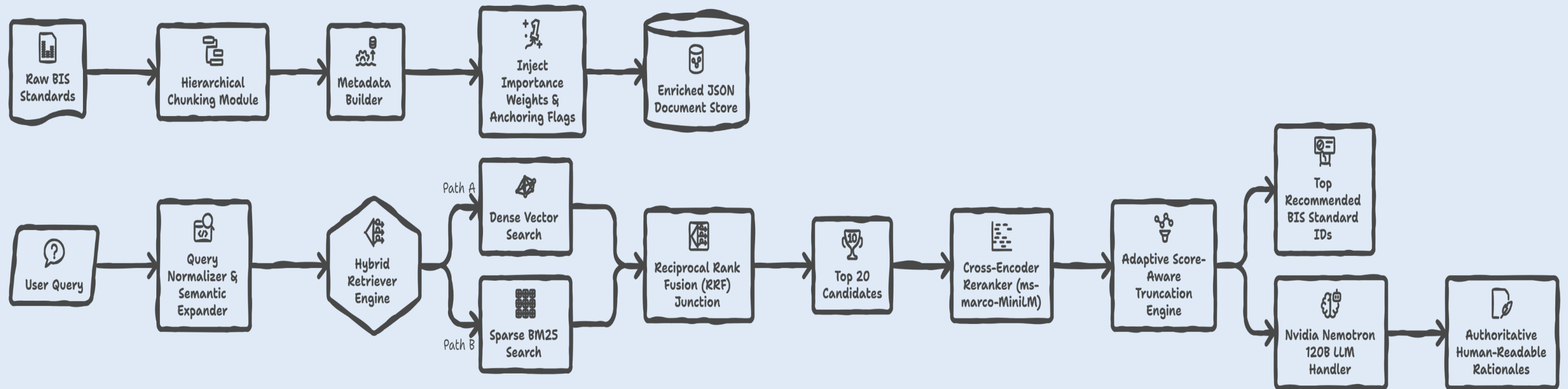
❖ Outcome :

- Transforms user queries into precise BIS standard recommendations.



❖ System Architecture:

Mission BIS - Hybrid RAG Recommendation Engine Workflow



❖ Chunking & Retrieval Strategy:

❖ Chunking Strategy

- Split BIS documents into **small meaningful parts (chunks)**
- Use **Document** → **Section** → **Clause structure**
- Create a **summary chunk (anchor)** for each standard
- Use **overlapping chunks** to avoid missing information
- Add metadata:
 - ❑ Category (cement, aggregate, etc.)
 - ❑ Keywords
 - ❑ Part number & year
- Improves accuracy and reduces noise

❖ Retrieval Strategy

- **Uses Hybrid Retrieval (Best of both worlds)**
- Uses both keyword and semantic search for better results
- BM25 (Keyword Search)
Finds exact terms like “33 grade cement”
Works well for technical and specific queries
- FAISS (Semantic Search)
Understands the meaning of the query
Finds relevant results even without exact words
- RRF (Fusion)
Combines results from both methods
Improves ranking and overall accuracy

❖ Demo Highlights:

BIS Compass
Standards discovery

+ New query

WORKSPACE

Recent

03:49 PM
Asbestos
Cement Sheets
IS 269:2015

Load

03:49 PM
Coarse & Fine
Aggregates
IS 269:2015

Load

03:49 PM
Specification
for precast...
IS 269:2015

Load

03:48 PM
Which
standard...
IS 269:2015

Load

03:48 PM
We
manufacture...
IS 269:2015

Load

Dark mode

BIS Compass
Compliance Intelligence Platform

Standards: 156 Latency: 0.9s Pipeline: 9

Discover **BIS Standards** Instantly

Describe your product or manufacturing process to find relevant compliance requirements powered by an intelligent RAG pipeline.

Describe your product to find applicable BIS standards... e.g., We manufacture 33 Grade Ordinary Portland Cement

Press **Enter** to search · **Shift + Enter** for newline

Find standards

33 Grade OPC Cement

Coarse & Fine Aggregates

Precast Concrete Pipes

Asbestos Cement Sheets

Portland Slag

TRY A SAMPLE QUERY:

We manufacture 33 Grade Ordinary Portland Cement. Which BIS standard applies?

Which standard applies to masonry cement for mortars, not structural concrete?

Standard for corrugated asbestos cement sheets for roofing and cladding?

Specification for precast concrete pipes with and without reinforcement for water mains.

Standard for Portland pozzolana cement that is calcined clay based.



BIS Compass
Standards discovery

+ New query

WORKSPACE

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BIS Compass
Compliance Intelligence Platform

Standards: 156 Latency: 0.9s Pipeline: 9

Specification for precast concrete pipes with and without reinforcement for water mains.

general

Find standards

33 Grade OPC Cement

Coarse & Fine Aggregates

Precast Concrete Pipes

Asbestos Cemen

#1 IS 269 : 2015 Best match 95%

Ordinary Portland Cement — Specification

IS 269 covers the specification for ordinary portland cement.

cement opc cement 33 grade portland opc

RATIONALE

IS 269 covers the specification for ordinary portland cement. Matches your query on: cement, 33 grade, portland.

RELATED STANDARDS

IS 8112 IS 12269 IS 4032

#2 IS 8112 : 2013 82%

Ordinary Portland Cement, 43 Grade — Specification

Matches your query's context of Portland cement manufacturing, though specifies 43 grade.

cement opc cement portland

Query Intelligence
Compact overview 0.87s

INPUT "Specification for precast concrete pipes with and..."

CATEGORY general

EXPANDED BIS standard compliance

PIPELINE STATS

20 Retrieved 10 Reranked 4 Output

Download results export
JSON snapshot (results + metadata) JSON

“Fast • Accurate • User-Friendly”

❖ Evaluation Results:

Performance Metrics

Metric	Value	Target
Hit Rate @3	100%	80%
MRR @5	0.95	0.7
Avg Latency	0.40 sec	5 sec

```
C:\Users\Anuja\Music\JAVA\mission BIS222\mission BIS>python eval_script.py --results results_uu.json
```

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BIS HACKATHON EVALUATION RESULTS

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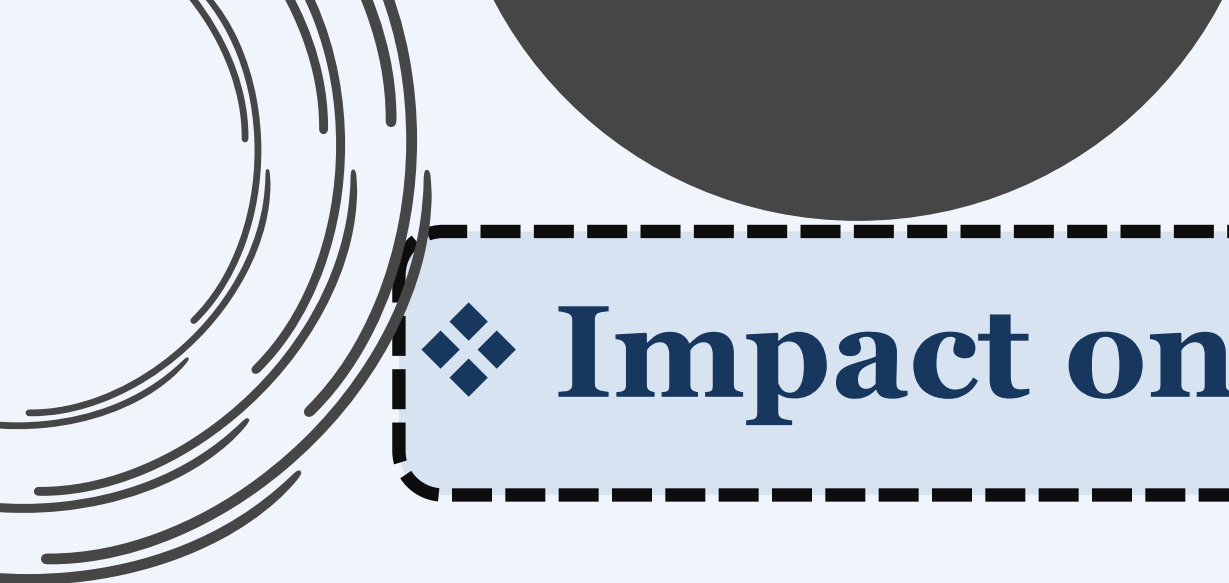
Total Queries Evaluated : 10

Hit Rate @3 : 100.00% (Target: >80%)

MRR @5 : 0.9500 (Target: >0.7)

Avg Latency : 0.40 sec (Target: <5 seconds)

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❖ **Impact on MSEs:**

❖ ***Faster Compliance***

Find relevant BIS standards in seconds
No manual document search

❖ ***Reduced Errors***

Accurate recommendations
Avoid wrong standard selection

❖ ***Cost Efficient***

No need for experts or consultants
Saves time and money

❖ ***Improved Decision-Making***

Clear insights for product planning
Faster, confident decisions

“Helps MSEs find the right BIS standards faster, cheaper, and with confidence”



❖ Team & Acknowledgements:

Name	Role
Yashraj A. Chavan	Lead AI Architect — RAG Pipeline, Retrieval System, LLM Integration
Rajvardhan R. Madekar	Frontend Developer — UI Design & User Experience
Aditya D. Magadum	Data Processing — Data Collection, Cleaning & Structuring
Shrivardhan R. Kedage	System Testing — Validation, Evaluation & QA

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- We acknowledge the **Indian Institute of Technology (IIT), Tirupati** for hosting and organizing this initiative.
- We are grateful for the **official BIS standards dataset and documentation** that enabled the development of our RAG-based recommendation system.
- We appreciate the use of **open-source tools, LLM frameworks, and retrieval technologies** that supported the implementation of our solution.